

Package (Library): These collection of Modules. Those are NUMPY,PANDAS,MATPLOTLIB,SEABORN

Modules: cillection of functions.

Functions: It is block of codes which performs specific code.

Input () Function: It takes information from the user

It allows user to ask question and user types the answer using keyboard.

```
In [1]: import math
x = sqrt(25) # This is name error where we have to import math functions before
            # So we dint get any answer from the code.
            # Here sqrt is inbuilt funtion.
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[1], line 2
      1 import math
----> 2 x = sqrt(25)

NameError: name 'sqrt' is not defined
```

```
In [4]: import math          # Here we import math module
x = math.sqrt(25)
x
```

Out[4]: 5.0

```
In [5]: x1 = math.sqrt(15)
x1
```

Out[5]: 3.872983346207417

Floor: It shows least or lower value.

Ceil: It shows maximum or highest value.

```
In [6]: import math
x1 = math.floor(2.3)
x1
```

Out[6]: 2

```
In [7]: import math as m    # we have given name to math as 'm' which we also call as a
x1 = m.floor(3.6)
x1
```

Out[7]: 3

```
In [8]: x1 = m.ceil(2.3)
x1
```

Out[8]: 3

```
In [9]: x1 = m.ceil(3.6) # 3 is smaller than 3.6
           # 4 is the next whole number above 3.6
           # ceil() moves the value upward to the nearest integer.
x1
```

Out[9]: 4

```
In [ ]: # Code Optimazation: It means combining all the codes together in one code.
```

```
In [10]: from math import *
print (math.floor(3.2))
print(math.ceil(3.6))
print(math.pow(3,2))
print(math.sqrt(35))
```

```
3
4
9.0
5.916079783099616
```

Input() Fuction: It takes information from user (question)and asks the user to enter the answer using keyboard.

It is in built funtion

```
In [13]: x = input()
y = input ()
z = x + y
print(z)  # here by default it take number as 'String' so addition is not possi
```

23

```
In [20]: x = input('Enter a Charecter')[3]
         print(x)
```

```
In [ ]: x = input('Enter a Charecter')[2]
         print(x)
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

Converting string to integer

```
In [15]: x = int(input('Enter the number'))    # Here we have converted 'String' to 'Int'
         y = int(input('Enter the number'))
         z = x + y
         print(z)
```

30

```
In [17]: a = int(input('Enter the number'))
         b = int(input('Enter the number'))
         c = a * b
         print(c)
```

120

Eval function using input: # eval() is used to evaluate a string as a Python expression.

When used with input(), it can automatically convert numbers and even calculate expressions.

input(): it takes text from the user

Eval(): converts the value and even calculates

Practice Questions

In [21]: *# Create a function to print your name.*

```
x = eval(input('enter the number'))
y = eval(input('enter the number'))
z = x* y
print(z)
```

30

In [22]: `name = input('enter the name')`
`print(name)`

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In [23]: *# Create a function to add two numbers.*

```
x = eval(input('enter the number'))
y = eval(input('enter the number'))
z = x + y
print(z)
```

33

In [24]: *# Create a function to find square of a number.*

```
import math as m
x = m.sqrt(16)
print(x)
```

4.0

In []: